OLA

Data Analyst Project

OLA Data Analyst Project

OLA Ride Booking Analysis

End-to-End Data Analytics Case Study

Prepared By

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Aspiring Data Analyst

Tools & Technologies

- **Microsoft Excel**
- **SQL (MySQL)**
- **Microsoft Power BI**

Project Details

Industry: Ride-Hailing & Transportation

Dataset: 20,000 Ride Booking Records

Analysis Period: July 2024

Project Link (Git-Hub Repo) : https://github.com/RushikeshTaralkar007/Data-Analytics-Projects-/tree/main

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Executive Summary

The OLA Ride Booking Analysis project was developed to understand how ride-booking data can be used to solve real business problems and support better decision-making. Using a dataset of over 20,000 ride bookings, I analyzed booking trends, revenue, customer behavior, vehicle performance, cancellation patterns, and customer satisfaction by using Microsoft Excel, SQL, and Power BI.

The project began with cleaning and preparing the raw data in Microsoft Excel to ensure accuracy and consistency. After the data was cleaned, SQL was used to answer important business questions, such as identifying successful bookings, analyzing revenue, tracking cancellations, and evaluating customer and driver ratings. These insights were then presented through an interactive Power BI dashboard that allows users to explore the data using filters, KPIs, and visualizations.

The dashboard provides a clear overview of the company's operational performance by highlighting key metrics such as total bookings, total revenue, cancellation rate, payment methods, vehicle performance, and customer ratings. It helps users quickly identify trends, compare performance, and understand the factors that influence business outcomes.

Working on this project gave me practical experience with the complete Data Analytics process, from data preparation and SQL analysis to dashboard development and business reporting. More importantly, it strengthened my ability to convert raw data into meaningful insights that can support business decisions.

Overall, this project demonstrates my understanding of end-to-end data analysis and reflects the technical and analytical skills required for a Data Analyst role. It has been developed as a portfolio project to showcase my ability to work with real-world business data and create solutions that provide valuable insights for stakeholders.

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Company Background

OLA is one of India's leading ride-hailing platforms, offering convenient and affordable transportation services across multiple cities. Through its mobile application, customers can book rides from different vehicle categories, while drivers can efficiently connect with passengers. Every day, the platform handles a large number of bookings, making data an essential part of its operations.

With thousands of rides taking place daily, OLA generates a significant amount of data related to bookings, revenue, customer preferences, payment methods, ride distances, cancellations, and service ratings. Analyzing this data helps the company understand customer behavior, monitor business performance, improve operational efficiency, and deliver a better user experience.

This project focuses on analyzing ride booking data to identify key trends and performance metrics that are important for business decision-making. By using Microsoft Excel, SQL, and Power BI, the project transforms raw booking data into meaningful insights through interactive dashboards and reports. These insights can help stakeholders monitor business performance, identify operational challenges, and make informed decisions to improve customer satisfaction and overall business growth.

This case study demonstrates how data analytics can be used to support business intelligence and highlights the value of turning operational data into actionable insights for better decision-making.

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Business Problem Statement

The ride-hailing industry operates in a fast-paced environment where thousands of bookings are processed every day. Managing such a large volume of operational data can be challenging without an effective analytics system. Business decisions based only on manual reports often make it difficult to identify trends, measure performance, and respond quickly to operational issues.

Some of the key challenges include understanding booking patterns, tracking revenue, identifying the reasons behind ride cancellations, evaluating customer and driver satisfaction, and monitoring the performance of different vehicle categories. Without proper analysis, these challenges can affect operational efficiency, customer experience, and overall business growth.

The objective of this project is to address these challenges by analyzing OLA ride booking data using Microsoft Excel, SQL, and Power BI. The project transforms raw booking data into interactive dashboards and meaningful business insights that help monitor key performance indicators (KPIs), identify trends, and support data-driven decision-making.

By providing a clear view of business performance, this analysis enables stakeholders to make informed decisions, improve operational efficiency, reduce ride cancellations, enhance customer satisfaction, and identify opportunities for revenue growth.

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Project Objectives

The primary objective of this project is to analyze OLA ride booking data and convert raw information into meaningful business insights using Microsoft Excel, SQL, and Power BI. By applying data analytics techniques, the project aims to understand business performance, identify operational trends, and support data-driven decision-making.

The specific objectives of this project are:

- Analyze booking trends and ride demand over the selected time period.
- Measure key business metrics such as total bookings, revenue, cancellation rate, and average booking value.
- Identify the most popular vehicle types based on booking volume and revenue.
- Evaluate customer and driver ratings to understand service quality and customer satisfaction.
- Analyze ride cancellation patterns and identify the major reasons behind cancellations.
- Compare different payment methods to understand customer payment preferences.
- Build an interactive Power BI dashboard with dynamic visualizations and KPIs.
- Generate actionable business insights and recommendations that can help improve operational efficiency and business performance.

This project demonstrates how data analytics can transform raw business data into valuable insights that support better decision-making and continuous business improvement.

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Dataset Overview

The dataset used in this project contains 20,000 ride booking records representing OLA's daily ride operations for the month of July 2024. It includes detailed information about each booking, such as booking status, customer details, vehicle type, ride distance, payment method, booking value, and customer and driver ratings. This dataset provides a comprehensive view of the ride-booking process and serves as the foundation for analyzing operational performance and customer behavior.

The data was organized in a structured format, making it suitable for cleaning, analysis, and visualization. Before performing any analysis, the dataset was reviewed to ensure consistency and accuracy. Data cleaning and preprocessing were carried out in Microsoft Excel, followed by SQL-based analysis and Power BI dashboard development.

The dataset was designed to answer important business questions related to booking trends, revenue generation, cancellation patterns, vehicle performance, customer satisfaction, and payment preferences. These insights help demonstrate how data analytics can support business decisions and improve operational efficiency.

Dataset Information :

Attribute	**Description**
Project Name	OLA Ride Booking Analysis
Industry	Ride-Hailing & Transportation
Dataset Size	20,000 Ride Booking Records
Analysis Period	July 2024
Data Format	Microsoft Excel (.xlsx)
Tools Used	Microsoft Excel, SQL, Power BI

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Key Data Fields

The dataset contains several important attributes used throughout the analysis, including:

- Date and Time
- Booking ID
- Booking Status
- Customer ID
- Vehicle Type
- Pickup & Drop Location
- Ride Distance
- Booking Value
- Payment Method
- Customer Rating
- Driver Rating
- Customer Cancellation Reason
- Driver Cancellation Reason
- Incomplete Ride Status

These fields were used to calculate key performance indicators (KPIs), perform SQL analysis, and build interactive dashboards that provide meaningful business insights and support data-driven decision-making.

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SQL Analysis

SQL played a key role in analyzing the OLA ride booking dataset and extracting meaningful business insights. After cleaning the dataset in Microsoft Excel, the data was imported into a SQL database where various queries were written to answer important business questions. These queries helped identify booking trends, customer behavior, revenue performance, cancellation patterns, and service quality.

To improve query organization and simplify reporting, SQL Views were created for each business requirement. This approach allows frequently used queries to be stored as reusable virtual tables, making the analysis more efficient and easier to maintain.

The following sections summarize the SQL queries used during the project.

Query 1: Retrieve All Successful Bookings

Objective: Identify all successfully completed rides.

SQL Operation Used:

```
Create view Succesful_Bookings As
select * From bookings where Booking_Status = 'Success';


select * From Succesful_Bookings;
```

Business Value:
This query filters only successful bookings, allowing the business to analyze completed rides, calculate revenue, and measure booking success rates.

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Query 2: Calculate Average Ride Distance by Vehicle Type

Objective: Compare the average ride distance for each vehicle category.

SQL Operation Used:

```
Create view ride_distance_for_each_vehicle As
select Vehicle_Type , Avg(Ride_Distance) AS Average_Distance From bookings group by
Vehicle_Type;

select * From ride_distance_for_each_vehicle;
```

Business Value:
The results help identify which vehicle types are commonly used for short or long-distance travel, supporting fleet planning and pricing strategies.

Query 3: Count Customer Cancelled Rides

Objective: Calculate the total number of rides cancelled by customers.

SQL Operation Used:

```
Create View canceled_ride_by_customers As
select Count(*) From bookings where Booking_Status = 'Canceled by Customer';

select * from canceled_ride_by_customers;
```

Business Value:
This analysis helps understand customer cancellation behavior and supports initiatives to reduce cancellation rates.

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Query 4: Top 5 Customers by Number of Bookings

Objective: Identify the most active customers.

SQL Operation Used:

```
Create View Top_5_Customers As
select Customer_ID , Count(Booking_ID) As
Total_Rides From bookings group by Customer_ID
order by Total_Rides desc Limit 5;

Select * From Top_5_Customers;
```

Business Value:
Recognizing frequent customers enables the business to design loyalty programs, personalized offers, and customer retention strategies.

Query 5: Driver Cancellations Due to Personal or Vehicle Issues

Objective: Count rides cancelled by drivers because of personal or vehicle-related reasons.

SQL Operation Used:

```
Create View Canceled_By_Driver_Personal_Issue As
select count(*) From bookings where Canceled_Rides_by_Driver = 'Personal & Car related
issue';

select * from Canceled_By_Driver_Personal_Issue;
```

Business Value:
This analysis helps identify operational issues affecting ride completion and supports improvements in driver management.

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Query 6: Maximum and Minimum Driver Ratings for Prime Sedan

Objective: Analyze driver performance for Prime Sedan rides.

SQL Operation Used:

```
Create View Min_Max_Driver_Ratings As
select max(Driver_Ratings) As Max_Ratings ,
min(Driver_Ratings) As Min_Ratings
From bookings where Vehicle_Type = 'Prime Sedan';

select * from Min_Max_Driver_Ratings;
```

Business Value:
The results highlight the range of driver ratings, helping evaluate service quality and identify opportunities for driver performance improvement.

Query 7: Retrieve All UPI Payment Bookings

Objective: Filter bookings completed using the UPI payment method.

SQL Operation Used:

```
Create View UPI_Payment As
select * From bookings where Payment_Method = 'UPI';

select * from UPI_Payment;
```

Business Value:
Understanding payment preferences helps the business evaluate digital payment adoption and design targeted promotional campaigns.

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Query 8: Average Customer Rating by Vehicle Type

Objective: Compare customer satisfaction across different vehicle categories.

SQL Operation Used:

```
create view customer_rating_per_vehicle_type As
select Vehicle_Type , Avg(Customer_Rating) As Avg_Customer_Ratings
From bookings group by Vehicle_Type;


Select * From customer_rating_per_vehicle_type;
```

Business Value:
This analysis identifies vehicle categories with the highest customer satisfaction and highlights areas that require service improvements.

Query 9: Total Booking Value of Successful Rides

Objective: Calculate total revenue generated from completed rides.

SQL Operation Used:

```
Create View Total_Success_Booking_Value As
select Sum(Booking_Value) As Total_Successful_Value
From bookings where Booking_Status = 'Success';

Select * From Total_Success_Booking_Value;
```

Business Value:
Revenue generated from successful bookings is a key business metric used to monitor financial performance and business growth.

Query 10: Retrieve Incomplete Rides with Reasons

Objective: Identify rides that were not completed and record the associated reasons.

SQL Operation Used:

```
Create View Incomplete_Rides_Along_With_Reason As
select Booking_ID ,Incomplete_Rides_Reason From bookings where Incomplete_Rides = 'Yes';

Select * from Incomplete_Rides_Along_With_Reason;
```

Business Value:
Analyzing incomplete rides helps identify operational bottlenecks and supports corrective actions to improve ride completion rates.

Summary

SQL was used to transform raw booking data into meaningful business information by filtering, aggregating, and organizing data into reusable views. The analysis supported the development of key performance indicators (KPIs) and provided valuable insights that were later visualized in the Power BI dashboard. These SQL queries formed the foundation for understanding booking performance, customer behavior, revenue generation, and operational efficiency throughout the project.

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Power BI Data Modeling & Dashboard Development

After completing data cleaning in Microsoft Excel and analysis in SQL, the processed dataset was imported into Power BI to build an interactive dashboard. The data was reviewed, transformed where required using Power Query, and organized into a structured model to ensure accurate reporting and efficient performance.

The dashboard was designed with a user-friendly layout, allowing stakeholders to explore business performance through interactive filters, slicers, KPI cards, and visualizations. The reports are divided into five analytical sections, each focusing on a specific business area.

Dashboard Structure

1. Overall Dashboard

- Ride Volume Over Time
- Booking Status Breakdown

This page provides a high-level overview of booking activity and overall business performance, helping users monitor booking trends and ride completion status.

2. Vehicle Analysis

- Top 5 Vehicle Types by Ride Distance
- Average Customer Rating by Vehicle Type

This section compares the performance of different vehicle categories, helping identify the most frequently used vehicles and evaluate customer satisfaction.

3. Revenue Analysis

- Revenue by Payment Method
- Top 5 Customers by Total Booking Value

- Ride Distance Distribution

The revenue dashboard focuses on financial performance by analyzing payment preferences, customer spending patterns, and ride distance trends that contribute to business revenue.

4. Cancellation Analysis

- Customer Cancellation Reasons
- Driver Cancellation Reasons

This page helps identify the major causes of ride cancellations, allowing the business to understand operational challenges and improve ride completion rates.

5. Ratings Dashboard

- Driver Rating Distribution
- Customer vs. Driver Ratings

The ratings dashboard evaluates service quality by comparing customer and driver ratings, helping measure customer satisfaction and driver performance.

Dashboard Features

The dashboard includes KPI cards, slicers, interactive charts, bookmarks, and navigation buttons, allowing users to filter data and explore insights dynamically. These features improve report usability and enable faster, data-driven decision-making.

Overall, the Power BI dashboard transforms raw booking data into meaningful visual insights, making it easier to monitor business performance, identify trends, and support strategic decision-making.

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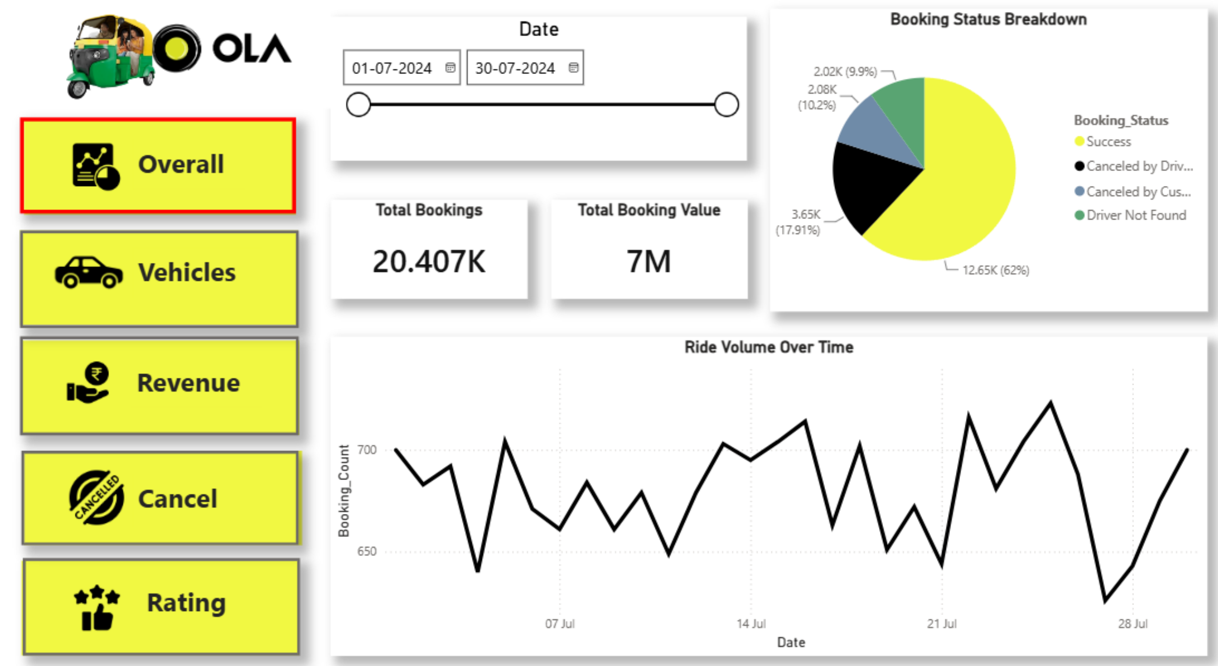
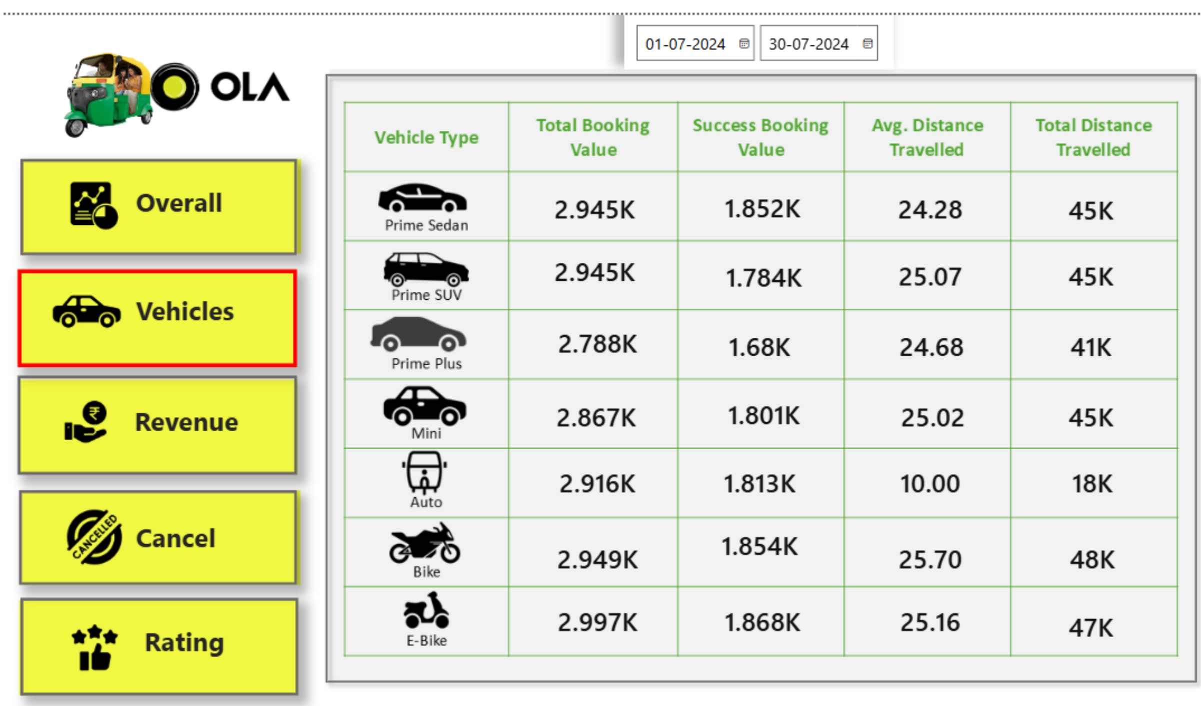
Overall Dashboard



Vehicle Dashboard



Vehicle Type	Total Booking Value	Success Booking Value	Avg. Distance Travelled	Total Distance Travelled
Prime Sedan	2.945K	1.852K	24.28	45K
Prime SUV	2.945K	1.784K	25.07	45K
Prime Plus	2.788K	1.68K	24.68	41K
Mini	2.867K	1.801K	25.02	45K
Auto	2.916K	1.813K	10.00	18K
Bike	2.949K	1.854K	25.70	48K
E-Bike	2.997K	1.868K	25.16	47K

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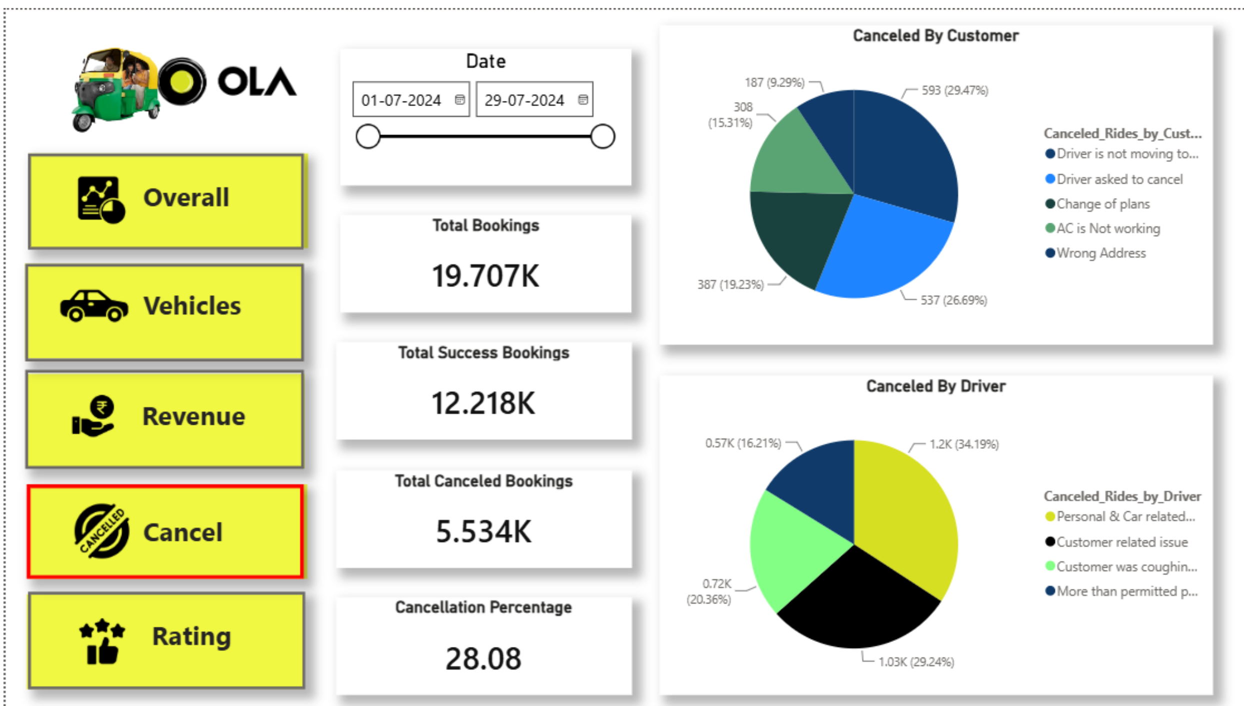
Revenue Dashboard

Cancellation Dashboard



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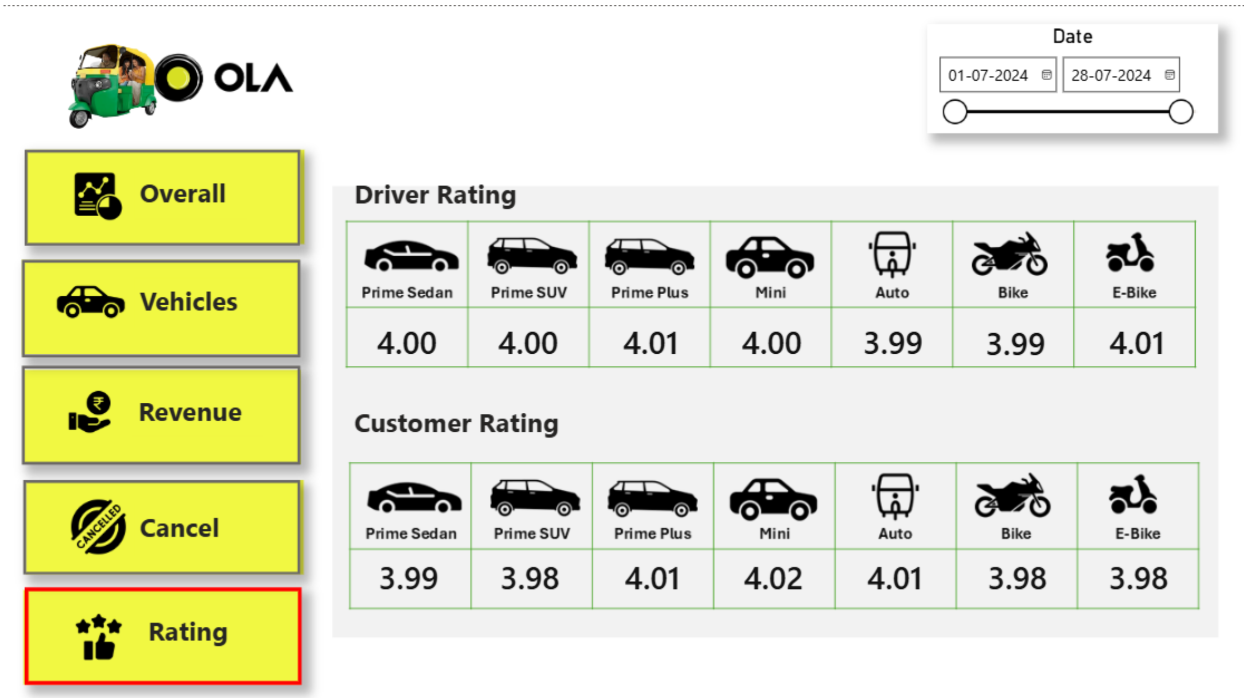
Ratings Dashboard



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Business Insights

The analysis of the OLA ride booking dataset revealed several valuable insights into booking performance, customer behavior, and operational efficiency. By combining SQL analysis with Power BI visualizations, meaningful patterns were identified that can support better business decisions.

Some of the key business insights are:

- A majority of ride bookings were completed successfully, indicating a stable overall service performance.
- Revenue was primarily generated through successful bookings, making ride completion a critical factor for business growth.
- Certain vehicle categories recorded higher booking volumes and longer ride distances, showing greater customer demand.
- Digital payment methods such as UPI were widely used, reflecting changing customer payment preferences.
- A small group of customers contributed significantly to the overall booking value, highlighting the importance of customer retention and loyalty programs.
- Customer and driver ratings remained consistently positive, indicating satisfactory service quality across most rides.
- Ride cancellations occurred due to both customer and driver-related reasons, suggesting opportunities to improve communication and operational planning.
- Booking demand varied throughout the analysis period, helping identify peak booking days and resource requirements.
- Comparing customer and driver ratings provided valuable insights into service quality and helped identify areas where customer experience could be improved.
- Interactive dashboards made it easier to monitor key performance indicators (KPIs), identify trends, and support faster, data-driven decision-making.

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Overall, the analysis demonstrates how business data can be transformed into actionable insights that help improve operational efficiency, customer satisfaction, and overall business performance. These findings can support management in making informed decisions and developing strategies for continuous improvement.

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Business Recommendations

Based on the analysis of the OLA ride booking data, several recommendations can help improve operational efficiency, customer satisfaction, and overall business performance.

- Reduce ride cancellations by improving driver allocation and minimizing customer waiting time.
- Monitor cancellation trends regularly to identify recurring issues and implement corrective actions.
- Encourage customers to use digital payment methods by offering rewards or cashback programs.
- Introduce loyalty programs for frequent customers to improve customer retention and increase repeat bookings.
- Reward high-performing drivers based on customer ratings to maintain service quality and encourage better performance.
- Optimize vehicle availability during peak demand periods to reduce waiting time and improve ride completion rates.
- Continuously monitor key performance indicators (KPIs) such as booking success rate, revenue, and customer ratings to evaluate business performance.
- Use booking trend analysis to improve demand forecasting and support better operational planning.
- Gather customer feedback regularly to understand service expectations and identify areas for improvement.
- Enhance the interactive dashboard by integrating real-time data, enabling management to make faster and more informed business decisions.

Implementing these recommendations can help improve customer experience, increase operational efficiency, reduce cancellations, and support sustainable business growth. The insights generated through this project demonstrate how data analytics can assist organizations in making informed, data-driven decisions and continuously improving their business performance.

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Challenges Faced

During the development of this project, several challenges were encountered while working with the dataset and building the dashboard. These challenges provided valuable learning opportunities and helped improve both technical and analytical skills.

One of the initial challenges was ensuring the quality of the dataset. The data required cleaning and validation to maintain consistency before analysis. Standardizing the data and preparing it for SQL and Power BI was an important step in achieving accurate results.

Another challenge was writing SQL queries to answer different business questions. Creating efficient queries using aggregate functions, grouping, filtering, and views required careful planning to ensure the results were both accurate and meaningful.

While developing the Power BI dashboard, organizing multiple visuals into a clean and interactive layout was another challenge. It was important to select appropriate charts, create meaningful KPIs, and design a dashboard that was easy to understand and navigate.

Interpreting the analytical results from a business perspective was also an essential part of the project. Instead of focusing only on technical outputs, the analysis was presented as actionable business insights and recommendations that could support better decision-making.

Overall, these challenges enhanced my understanding of the complete Data Analytics workflow and strengthened my skills in data preparation, SQL analysis, dashboard development, and business reporting.

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Learning Outcomes

This project provided valuable hands-on experience in the complete Data Analytics workflow, from data preparation to business reporting. Working with a real-world ride booking dataset helped me strengthen both my technical skills and my ability to solve business problems using data.

Throughout the project, I improved my knowledge of Microsoft Excel by cleaning and preparing raw data for analysis. Using SQL, I gained practical experience in writing queries to answer business questions, perform data aggregation, and extract meaningful insights. I also enhanced my Power BI skills by creating interactive dashboards, developing KPIs, and designing visual reports that effectively communicate business performance.

Beyond technical skills, this project improved my analytical thinking and problem-solving abilities. I learned how to interpret data from a business perspective, identify trends, and present findings in a clear and meaningful way. It also strengthened my understanding of data storytelling and the importance of converting raw data into actionable insights that support business decisions.

Overall, this project has increased my confidence in working with data and provided practical experience that aligns with the responsibilities of a Data Analyst. It has also helped me build a strong portfolio project that demonstrates my technical knowledge, business understanding, and ability to deliver data-driven solutions.

Future Enhancements

While this project provides valuable insights into OLA ride booking operations, there are several opportunities to further enhance its functionality and business value.

One potential improvement is integrating Python to automate data cleaning, preprocessing, and advanced data analysis. This would reduce manual effort and improve the efficiency of handling larger datasets. The dashboard can also be connected to Power BI Service to enable automatic data refresh and real-time reporting for business users.

Future versions of the project could include predictive analytics to forecast ride demand, estimate revenue trends, and identify peak booking periods. Machine learning models could also be used to predict ride cancellations and customer behavior, helping the business make proactive decisions.

Additional enhancements such as interactive drill-through reports, advanced KPI tracking, and integration with cloud databases like Azure SQL or MySQL would make the solution more scalable and suitable for enterprise-level reporting.

Overall, these enhancements would transform the project from a descriptive analytics solution into a more advanced Business Intelligence platform capable of supporting real-time monitoring and predictive decision-making.

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Conclusion

The OLA Ride Booking Analysis project successfully demonstrates the complete Data Analytics process, from data preparation and SQL analysis to interactive dashboard development using Power BI. By analyzing over 20,000 ride booking records, the project provided meaningful insights into booking trends, revenue performance, customer behavior, vehicle utilization, and ride cancellations.

Through this project, I gained practical experience in cleaning and analyzing data, writing SQL queries, building interactive dashboards, and presenting business insights in a clear and meaningful way. The dashboard enables users to monitor key performance indicators (KPIs), identify business trends, and make informed decisions based on data.

More importantly, this project helped me understand how data analytics can solve real business problems by converting raw data into actionable insights. It strengthened both my technical skills and my ability to think from a business perspective while working with real-world datasets.

Overall, this project reflects my ability to apply Microsoft Excel, SQL, and Power BI to develop an end-to-end Business Intelligence solution. It serves as a strong portfolio project that demonstrates my analytical thinking, problem-solving skills, and readiness to contribute as a Data Analyst in a professional environment.